

OUTPUT: File Integrity Checker

1. Initial hashes need to be created for all the files of the Testing_Directory (This step should be done only once).

```
# Step 1: Initialize hashes (run this only once)
InitialHashes(monitored_directory)
```

```
PS C:\Users\surat\OneDrive\Desktop\Task1> & C:/Users/surat/OneDrive/Desktop/Task1/file_integrity_checker.py
[✓] No modifications detected. All files are intact.
PS C:\Users\surat\OneDrive\Desktop\Task1>
```

2. Modifications made in the files of the directory.

```
Testing_Directory > file1.txt
1 This is test file 1 and modified !

Testing_Directory > file2.txt
1 This is test file 2 and modified!

Testing_Directory > file3.txt
1 This is test file 3 and modified!

Testing_Directory > file4.txt
1 This is test file 4 and modified!

Testing_Directory > file5.txt
1 This is test file 5 and modified!
```

- Comment this out and run the integrity check again

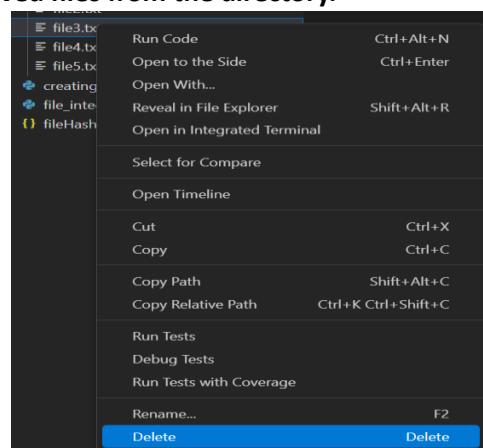
```
# Step 1: Initialize hashes (run this only once)
# InitialHashes(monitored_directory)
```

- Changes in the files detected

```
PS C:\Users\surat\OneDrive\Desktop\Task1> & C:/Users/surat/OneDrive/Desktop/Task1/file_integrity_checker.py
🚨 Integrity Alert! Detected changes in file

Modified Files:
- Testing_Directory\file1.txt
- Testing_Directory\file2.txt
- Testing_Directory\file3.txt
- Testing_Directory\file4.txt
- Testing_Directory\file5.txt
PS C:\Users\surat\OneDrive\Desktop\Task1>
```

3. Deleting one of the saved files from the directory.



file3.txt has been deleted

- Deletion of the file has been detected

```
PS C:\Users\surat\OneDrive\Desktop\Task1> & C:/Users/surat/OneDrive/Desktop/Task1/file_integrity_checker.py  
  
🔥 Integrity Alert! Detected changes in files:  
  
⚠️ Missing Files:  
- Testing_Directory\file3.txt  
PS C:\Users\surat\OneDrive\Desktop\Task1>
```

4. Creating a new file viz file6.txt in the directory.

```
Testing_Directory  
Testing_Directory > file6.txt  
1 This is a new file!
```

- New file is detected!

```
PS C:\Users\surat\OneDrive\Desktop\Task1> & C:/Users/surat/OneDrive/Desktop/Task1/file_integrity_checker.py  
  
🔥 Integrity Alert! Detected changes in files:  
  
NEW New Files Detected:  
- Testing_Directory\file6.txt  
PS C:\Users\surat\OneDrive\Desktop\Task1>
```